CURVES DEFINED PARAMETRICALLY **03/03/26**

1. By finding some points, sketch the curve given by $x = \sin 2t$, $y = \sin t$ for $0 \le t \le 2\pi$.

2. Find a Cartesian equation for the curve defined by $x = \frac{1+t}{1-t}$, $y = \frac{2+t}{1-t}$.

3. A curve is defined by $x = -2 + 3t$, $y = 1 - 2t$ for $t \in \mathbb{R}$. Find the points corresponding to $t = -2, -1, 0, 1, 2$. What sort of curve is it? Find its Cartesian equation.

4. By finding points corresponding to $t = -2, -1, 0, 1, 2$,
   sketch the curve defined by the parametric equations $x = -2 + 3t^2$, $y = 1 - 2t^2$.

   Find the Cartesian equation of the curve. Compare to question 1.

5. By finding points corresponding to unit values of t, sketch the curve with parametric equations $x = 2t - 1$, $y = t^2$ for $-4 \le t \le 4$. Confirm your result by finding the curve's Cartesian equation.

Intersections.

6. Find the points of intersection of the line $y = 2x + 1$ with the curve $x = t^3, y = t^2 + 2t$.

7. Find the points of intersection of the parabola $x + y^2 = 9$ and the curve $x = (t-3)^2, y = 2t$.

8. The variable point P moves as a parameter t varies: the coordinates of P are $(t^2, 2t)$.
   Find the points where the locus of P crosses the straight line $y = 2x - 4$.

9. The curve $x = t + 1$, $y = t^2 - k$ intersects the x-axis at P and Q (where P is to the left of Q), and the y-axis at the point R.
   Find the value of k ($k \ne 1$) such that $OQ = 2OR$.

10. In each of the following:

    (a) try to sketch the curve;
    (b) find the Cartesian equation of the curve, explaining whether all values of x and y satisfying this equation lie on the curve you have drawn.

    i. $x = \sin^2\theta, y = 1 + 2\sin\theta$
    ii. $x = 2\cos\theta, y = 2\sin\theta$
    iii. $x = 3\cos\theta, y = 2\sin\theta$
    iv. $x = 4 + 3\cos\theta, y = 1 + 3\sin\theta$
    v. $x = \dfrac{t}{1+t}, y = \dfrac{t}{1-t}$
    vi. $x = t, y = \dfrac{1}{t}$
    vii. $x = 2^t, y = 2^{-t}$

    Explain why the set of points described in vii is a strict subset of the set of points described in vi.